airBP: Monitor Your Blood Pressure with Millimeter-Wave in the Air

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Blood pressure (BP), an important vital sign to assess human health, is expected to be monitored conveniently. The existing BP monitoring methods, either traditional cuff based or newly emerging wearable based, all require skin contact, which may cause unpleasant user experience and is even injurious to certain users. In this article, we explore contactless BP monitoring and propose airBP, which emits millimeter-wave signals toward a user's wrist, and captures the reflected signal bounded off from the pulsating artery underlying the wrist. By analyzing the reflected signal strength of the signal, airBP generates the arterial pulse and further estimates BP by exploiting the relationship between the arterial pulse and BP. To realize airBP, we design a new beam-forming method to keep focusing on the tiny and hidden wrist artery, by leveraging the inherent periodicity of the arterial pulse. Moreover, we custom design a pre-training and neural network architecture, to combat the challenges from the arterial pulse sparsity and ambiguity, so as to estimate BP accurately. We prototype airBP using a coin-size commercial off-the-shelf millimeter-wave radar and perform extensive experiments on 41 subjects. The results demonstrate that airBP accurately estimates systolic and diastolic BP, with a mean error of −0.30 mmHg and −0.23 mmHg, as well as a standard deviation error of 4.80 mmHg and 3.79 mmHg (within the acceptable range regulated by the FDA's AAMI protocol), respectively, at a distance up to 26 cm.

CCS Concepts: • **Human-centered computing → Ubiquitous and mobile computing design and evaluation methods;**

Additional Key Words and Phrases: Blood pressure, contactless sensing, millimeter-wave sensing

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1 INTRODUCTION

Recently, we have witnessed a surge of contactless human sensing research in the IoT community using wireless signals [1, 2]. Compared to WiFi, UWB, and other wireless signals, **Millimeter-Wave (mmWave)** sensing attracts attention due to its advantages of high-resolution sensing ability [3–5]. By extracting and analyzing mmWave signals reflected from the human body, one can derive a series of human information in a contactless manner: human activity recognition [6, 7], respiration and heart rate monitoring [8–10], and even the subtle and intricate cardiac activities, such as SCG (seismocardiography) and ECG (electrocardiogram) recovery [3, 5, 11]. Yet, another very important vital sign, **Blood Pressure (BP)**, remains a hard open problem to be sensed in a contactless way.

As a key metric to assess human health, low BP indicates many terrible risks, such as dehydration, anemia, and heart diseases, whereas high BP indicates hypertension and potential cerebrovascular accidents [12–14]. Currently, about 31.1% of adults are affected by hypertension and need frequent BP monitoring for awareness and control of BP levels [15]. To obtain BP, traditional cuff-based methods employ an inflatable cuff to wrap a user's arm, and then inflate and deflate the cuff successively to detect the pulse sound of the brachial artery, so as to get BP values according to the cuff pressure [14]. Although they provide reliable performance, these methods usually require bulky and cumbersome devices and thus interfere with users' normal activities. For more convenient BP monitoring, eBP [16] builds an in-ear device that integrates a tiny pump, an inflatable balloon, and a PPG (photoplethysmography) sensor to monitor BP from the users' ear. LiveOne [17] also proposes wrist-worn devices with skin-close-fitting pressure sensors to measure BP. However, the compression still causes an unpleasant user experience and is even injurious (e.g., tissue hypoxia) to certain users who require frequent measurements [18]. Thus, BP monitoring using cuffless devices rises along with the proliferation of smartphones and smartwatches. For instance, Seismo [19] utilizes a smartphone's built-in accelerator and camera to measure the **Pulse Transit Time (PTT)** between one's heart and finger, and maps the PTT to BP; SeismoWatch and Crisp-BP [20, 21] alternatively adopt a smartwatch for the task. Despite remarkable progress, we note that these works all require *direct skin contact*, imposing and maintaining an appropriate and consistent contact pressure on the skin, which itself is a difficult task [21]. The strict contact requirement hinders comfortable BP monitoring and widespread usage in daily life, which is not suitable particularly for the elderly, children, or in-sleep users.

In this article, we envision *contactless* BP sensing. As illustrated in Figure 1, a user's BP can be captured by a sensing gadget at a distance, without wearing any sensor. Such a contactless way brings significant benefits of zero disturbance—that is, BP monitoring can be accomplished without awareness, whenever the user is working or sleeping, and so forth. To realize the vision, we develop a system called *airBP*, which emits mmWave signals toward a user's wrist and captures the reflected signals bounded off from the pulsating arteries underlying the wrist. By analyzing the signal, airBP recovers the pulse waveform of arterial activities, from which airBP extracts the BP-related features (i.e., the **Reflected Wave Transit Time (RWTT)**) and further estimates BP.

To enable precise BP estimation while delivering a comfortable user experience, airBP faces many challenges that have not been well solved by existing methods. Foremost, the feasibility of the first step of airBP—that is, "in-air" detecting of the minute pulse of the tiny wrist artery at a distance (i.e., 12 cm)—has not been confirmed. As shown in Table 1, the size and extension amplitude of the tiny artery is only $\sim$ 3 mm and $\sim$ 0.03 mm [22], which is merely 2% and 6% of that in respiration [23] and heart [24] rate sensing. The most recent work [25] takes initial steps to sense the wrist artery pulse via mmWave radar, but it operates at rather close distances (i.e. 3 mm) and is accurate in very limited cases (i.e., only 5% of test samples). Our work aims to push the boundary

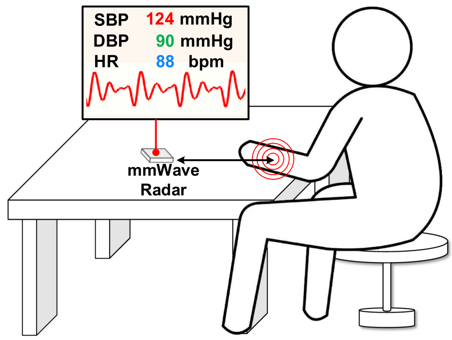

Fig. 1. An illustration of contactless BP monitoring via mmWave sensing.

Table 1. Comparison of Sensing Targets and Vibration Amplitude

Sensing Task	Target	Diameter	Amplitude
Respiration	Chest	253 mm [27]	12.6 mm [23]
Heartbeat	Heart	121 mm	5 mm [24]
Blood Pressure	Wrist Artery	3 mm	0.03 mm [22]

of contactless BP inference at farther distances (e.g., 12 cm) by uncovering the underlying interaction between the mmWave signal's variation and wrist arterial activity. Specifically, we perform a series of experiments using a custom-built testbed consisting of an mmWave radar, artificial blood vessels, and arms driven by a reconfigurable peristaltic pump. We have two major findings. First, the mmWave signal's phase variation, which is commonly leveraged by existing works [3–5] for micro-vibration or cardiac activity, does not apply. An in-depth investigation reveals the root reason: the impact of skin and muscles that closely wrap the blood vessels. Second, in contrast, there is a strong correlation between blood volume in the wrist artery and the **Received Signal Strength (RSS)**. We model the correlation using the Lambert-Beer law [26], which validates the efficacy of mmWave sensing for the wrist artery pulse.

Then, to drive airBP from feasibility to practice, we have to address two major challenges: focusing on the tiny and hidden wrist artery and learning BP directly from the arterial pulse waveform.

Focusing on the Tiny and Hidden Wrist Artery. To recover a *stable and accurate* waveform of the minute pulse on a tiny wrist artery is quite hard, particularly considering that the artery is hidden under skin and tissues, and is extremely sensitive to even slight arm movement. Although existing methods can compress sensors right on the wrist artery [17, 21, 28] to obtain high-quality arterial pulses, airBP, as a contactless method operating at a farther distance, does *not* have such privilege. A straightforward solution may apply beam-forming—that is, scanning the whole concerned area, and denoting the sub-area with the maximum RSS as the target [29]. However, we find that such an RSS-maximizing approach does not fit, which usually focuses on other tissues (e.g., arm muscles) with stronger reflectivity rather than the subcutaneous artery. In airBP, we propose a new **Autocorrelation Coefficient (ACC)**-maximizing beam-forming, which leverages the inherent periodicity of arterial pulse: a time series of periodical signals itself has a large ACC value. Thus, airBP keeps computing ACC during beam-forming, and focuses on the sub-area identified with the maximum ACC, which leads to stable and accurate arterial pulse recovery.

Learning BP Directly from the Arterial Pulse Waveform. Once deriving pulse waveforms, previous contact-based works distill BP in two stages: first computing BP-related features (i.e., RWTT) from the waveform and then using supervised learning to map the RWTT to BP [21]. However, it fails for contactless airBP to accurately extract RWTT: the ambiguous pulse waves sensed from a distance (e.g., 12 cm) are of low signal-to-noise ratio, which makes it difficult to accurately capture the minor arterial diameter changes caused by reflected blood flow. In this work, we let airBP learn BP directly from the pulse waveform. To enable this, we customize the state-of-the-art pre-training and transformer learning methods to circumvent the adverse "sparsity" of the arterial pulse waveform. In particular, different from image and speech where information is redundantly dense, the distribution of BP-related points in a pulse waveform is extremely *sparse*—that is, specific key points are separated by a long and uncertain series of data. Thus, learning such sparse features is quite challenging, particularly considering that the ground-truth label (i.e., only a numerical BP value) provides rather coarse and insufficient weak supervision. To handle the problem, we design *BP-aware pre-training*, which pre-trains airBP with stronger supervision (i.e., long-term representation of BP-related data points). The pre-training boosts airBP's awareness of sparse BP-related data and helps discriminate BP features from the other irrelevant features. In addition, for accurate BP estimation, we employ a convolutional transformer with a *multi-level attention mechanism*. This not only enables global attention to enable the extraction of long-term BP-related features from the whole waveform series regardless of separation but also leverages conventional self-attention layers to exploit the inherent periodicity of the BP features, so as to increase BP estimation accuracy.

We implement airBP using a coin-size COTS (commercial off-the-shelf) mmWave radar [30] and evaluate its performance on 41 subjects. Extensive experimental results show that airBP provides precise estimation for both **Systolic Blood Pressure (SBP)** and **Diastolic Blood Pressure (DBP)**, with a **Mean Error (ME)** of −0.30 mmHg and −0.23 mmHg, as well as a **Standard Deviation (STD)** of 4.80 mmHg and 3.79 mmHg, respectively. It meets the standard of the AAMI (Association for the Advancement of Medical Instrumentation) [31], which accepts ME within 5 mmHg and STD within 8 mmHg. Meanwhile, experimental results in different scenarios verify that airBP has the following advantages: (i) *flexibility*, with an effective range of up to 26 cm; (ii) *robustness*, which is effective regardless of ambient light conditions; and *universal*, which is valid in unseen scenarios surrounded by clutter.

Our key contributions can be summarized as follows:

— We explore and validate the feasibility of contactless BP monitoring by uncovering the strong correlation between arterial activity and RSS changes even on tiny and hidden wrist arteries (Section 2).
— We propose an ACC-maximizing beam-forming method to focus on arterial pulse sensing (Section 4), as well as a custom-designed deep learning pipeline to learn BP directly from the sparse pulse waveform (Section 5).
— We prototype a contactless BP monitoring system, named *airBP*, with a coin-size COTS mmWave radar. Extensive experimental results on 41 subjects verify airBP's accurate and robust BP estimation with an effective range up to 26 cm (Section 7).

2 UNDERSTANDING BP MEASUREMENT VIA MMWAVE SENSING

At first glance, BP measurement from an arterial pulse should consist of three steps [21]: (i) sensing wrist artery activities and recovering the arterial pulse waveform, (ii) computing BP-related features from the waveform (see Appendix A for details), and (iii) obtaining BP using the extracted features. In the following, we try these steps using mmWave sensing. Specifically, we first present the preliminaries of mmWave sensing in Section 2.1, verify the feasibility of wrist artery sensing in Section 2.2, and finally figure out challenges in estimating BP via mmWave sensing results in Section 2.3.

2.1 Preliminaries of mmWave Sensing

The core idea to sense the arterial activities via mmWave is to transmit mmWave signals toward the wrist artery and analyze the reflected signals that are received by the receiving antenna.

The transmitted signal $x_{Tx}(\tau)$ and received signal $x_{Rx}(\tau)$ of **Frequency-Modulated Continuous Wave (FMCW)** mmWave radar at time τ within a chirp period T_s could be formulated as follows:

$$x_{Tx}(\tau) = S_{Tx} \cdot e^{j(2\pi f_c \tau + \frac{\pi B}{T_s}\tau^2)}, \tag{1}$$

$$x_{Rx}(\tau) = S_{Rx} \cdot e^{j(2\pi f_c(\tau - t_d) + \frac{\pi B}{T_s}(\tau - t_d)^2)}, \tag{2}$$

where S_{Tx} and S_{Rx} denote the signal strength (amplitude) of transmitted and received signals, f_c is the starting frequency, B is the bandwidth, and t_d is the round-trip time delay between transmitting and receiving. The FMCW radar further mixes $x_{Tx}(\tau)$ and $x_{Rx}(\tau)$ and outputs the intermediate frequency signal $x(\tau)$ as follows:

$$x(\tau) = x_{Rx}(\tau)x_{Tx}(\tau) \approx S_{Tx}S_{Rx} \cdot e^{j(\frac{4\pi Bd}{cT_s}\cdot\tau + \frac{4\pi f_c d}{c})}, \tag{3}$$

where d is the distance between the radar and reflecting objects, and c denotes the light speed.

In practice, $x(\tau)$ contains the reflection of different objects from various distances. So range-FFT [44] is used to separate these reflections from different distances based on their frequency components $\frac{4\pi Bd}{cT_s}$. When performing range-FFT, $x(\tau)$ is first sampled at discrete time intervals, defined as $x[\tau_n]$, where $n = 0, 1, \ldots, N_\tau - 1$, and N_τ is the number of samples per chirp. Then the range-FFT (implemented as Discrete Fourier Transform $\mathscr{F}$) is performed on $x[\tau_n]$ to obtain frequency components $X[k]$, where $k = 0, 1, \ldots, N_\tau - 1$:

$$X[k] = \mathscr{F}[x[\tau_n]] = \sum_{n=0}^{N_\tau - 1} x[\tau_n] \cdot e^{-j2\pi \frac{kn}{N_\tau}}. \tag{4}$$

Each frequency component $X[k]$ corresponds to the reflection from the range within $[\frac{kc}{2B}, \frac{(k+1)c}{2B}]$. Then the desired signal of the wrist artery X_{artery} is selected from the range bins (d_{artery}) where the wrist artery lies:

$$X_{artery} = X[d_{artery}]. \tag{5}$$

Finally, the variations of X_{artery} over slow time t are captured, which is denoted as $X_{artery}(t)$. (Note that $t \gg T_s$, during which multiple chirps are captured.) $X_{artery}(t)$ could be represented as follows:

$$X_{artery}(t) = S(t) \cdot e^{j\phi(t)}, \tag{6}$$

where $S(t)$ and $\phi(t)$ are the RSS and phase variation over time. Note that the distances changes caused by the wrist artery is only about 0.03 mm [22], which is rather smaller compared to the resolution of range bins $\frac{c}{2B}$ (about several centimeters), so the range bin of the wrist artery (d_{artery}) could be approximated as constant when extracting $X_{artery}(t)$.

It is expected that from changes of certain signal parameters (i.e., RSS $S(t)$ and phase $\phi(t)$ in Equation (6)), we may recover the periodic arterial pulse waveform. In the following, we conduct experiments to verify the feasibility.

2.2 Feasibility of Wrist Artery Sensing with mmWave in the Air

Although previous works have leveraged mmWave to obtain heart activities [3, 10, 11], it is not clear whether the weaker wrist artery motion, whose pulsating amplitude is only 6% of the heart as shown in Table 1, can be sensed at a distance (e.g., 12 cm). Thus, we start the feasibility exploration using a setup shown in Figure 2(a). We ask 41 subjects to separately place his/her wrist 12 cm upon

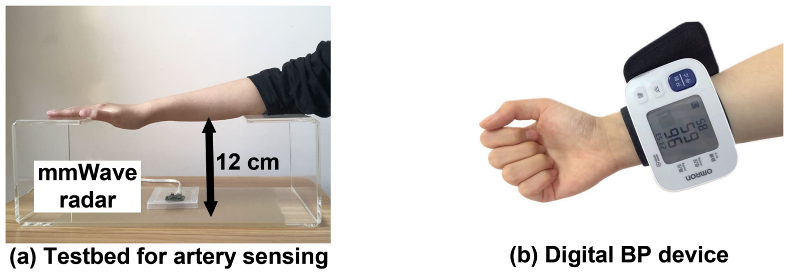

Fig. 2. Setup for our contactless BP monitor (a) and the device for ground-truth BP (b).

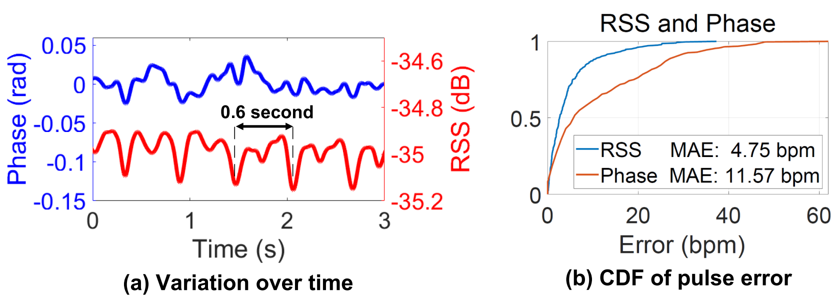

Fig. 3. Phase vs. RSS variation in wrist artery sensing: waveforms (a) and CDF of pulse rate estimation errors (b). Compared to RSS (red line in (a)), phase variation (blue line in (a)) generates pulse waveforms of low quality, which leads to inaccurate sensing results of arterial pulse as shown in (b).

Soli, a COTS FMCW mmWave radar operating at 60 GHz [30], for about 35 seconds. Then we utilize Soli to transmit and receive the reflected signal bounded off the wrist. For each experiment, we also get the ground-truth pulse rate using Omron, an FDA-approved cuff-based device [32] as shown in Figure 2(b). In total, 287 pulse waveforms from 41 subjects are collected.

Phase vs. RSS. We first use phase $\phi(t)$ to infer pulse waveforms, as it is widely used to sense minor motions like human chest vibrations caused by heart activity [3] and micrometer-level machine vibrations [4]. One representative result is plotted as the blue line in Figure 3(a), which is far from satisfactory (i.e., exhibiting poor periodicity). Quantitatively, we compute the absolute errors between the recovered pulse rates and the ground truth from all samples, and plot the CDF in Figure 3(b). We find that the recovered values deviate much from the ground truth with a high **Mean Absolute Error (MAE)** of 11.57 bpm. We then employ an alternative parameter of the signal (i.e., RSS $S(t)$) and repeat the preceding procedure. The result is plotted as the red lines in Figure 3(a), from which RSS leads to a much smaller MAE of 4.75 bpm as shown in Figure 3(b) by blue lines. This implies that RSS is more promising for wrist artery pulse sensing, which is different from previous common thoughts.

Why Is RSS better? We further conduct progressive in vitro experiments to examine the underlying reason. First, we build an in vitro blood circulation simulator consisting of an artificial blood vessel and a peristaltic pump to simulate a human heart's periodical blood pumping, as shown in Figure 4(a). The pump periodically transports blood at a frequency of 1.5 Hz (i.e., simulating a heart rate of 90 bpm). We then use the same RSS and phase-based algorithms as previously for pulse waveform recovery and plot the results in Figure 4(a). We can observe that both algorithms successfully recover the periodical pulse waveform, with the exactly accurate heart rate estimation (i.e., 88.9 bpm). Second, we simulate blood flow in blood vessels warped by muscles and skin using

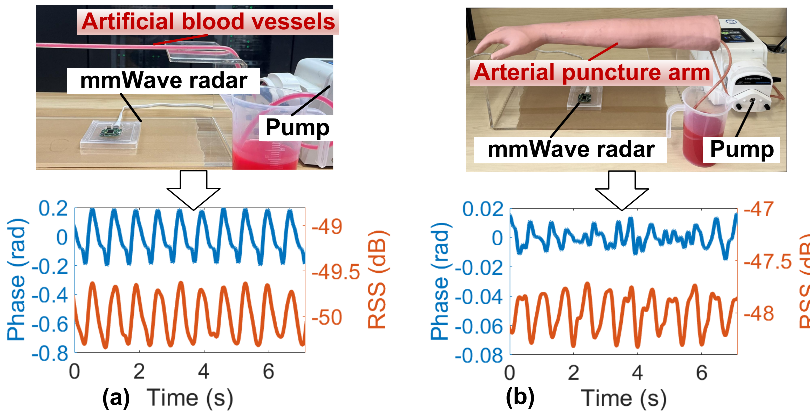

Fig. 4. Sensing the arterial pulse in an artificial blood vessel (a) and an arterial puncture arm (b). Under the hindrance of skin, RSS changes could still capture minor artery activities but phase variation could not.

an artificial arm that features the realism of actual arterial in medical training [33]. We perform the same experiments, as shown in Figure 4(b). From the results, the phase-based algorithm becomes ineffective and results in poor pulse waveforms, with poor periodicity and inaccurate heart rate estimation of 115 bpm. In contrast, the RSS-based algorithm keeps perfect recovery with an accurate heart rate estimation of 90.6 bpm. The results show that under the hindrance of skin and muscles, the minor distension of the wrist artery is difficult to capture via mmWave's phase changes, but it can be captured by RSS changes.

Formulating the Interaction between the mmWave Signal's Variation and Arterial Activity. To better understand the experimental results, we now give theoretical formulation. We first model the structure of the radial artery as illustrated in Figure 5(a), which is about 3 mm deep under the skin [34]. When the mmWave signals transmitted by the radar penetrates the skin and reaches the blood vessel [35], according to the Fresnel equation [36], part of them will be reflected at the interface (i.e., the surface of the skin and the blood vessel) and be received by the receiving antenna.

Assuming that the subject keeps the arm absolutely still, the variation of the reflected mmWave signal $X_{artery}(t)$ is only related to that of arterial activity. Since phase changes $\Delta\phi(t)$ are associated with the distance variation between radar and the target [37], one could obtain the diameter changes of blood vessels $\Delta L(t)$ based on the phase changes $\Delta\phi(t)$ of signals reflected by the upper surface of blood vessels (denoted by E_1 in Figure 5(a)) as follows:

$$\Delta\phi(t) = 2\pi\frac{\Delta L(t)}{\lambda}, \tag{7}$$

where λ denotes the mmWave signal's wavelength. Results shown in Figure 4 verify that such a method works well to recover the blood vessels' activity without the hindrance of skin. However, in noisy environments under the skin, it does not apply.

Luckily, besides the distance variation caused by blood vessels, the arterial activity also brings in the volume change of the blood flow, as shown in the bottom part of Figure 5(b). We now formulate its unique impact on mmWave signals. According to the Lambert-Beer law [26, 38], mmWave signals reflected by the bottom surface of the artery (denoted by E_2 in Figure 5(a)) is partly absorbed when traveling through the blood flow. The absorption coefficient of the blood flow $A_{blood}(t)$ could be formulated as follows:

$$A_{blood}(t) = e^{-\varepsilon(\lambda)\hat{c}\cdot 2L(t)}, \tag{8}$$

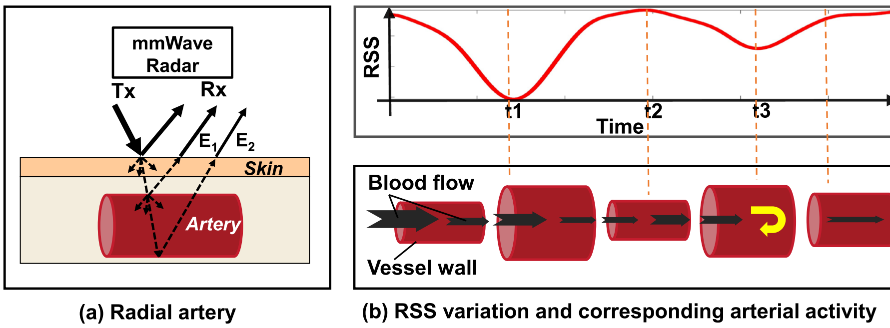

Fig. 5. An illustration of the radial artery (a) and the interaction between RSS change and arterial activity (b).

where $\varepsilon(\lambda)$ denotes the absorption coefficient for the wireless signal of wavelength λ, $\hat{c}$ is the concentration of the substance absorbing the signal in the blood flow, and $2L(t)$ denotes the signal's path length in the blood (i.e., twice the diameter of blood vessels at time t). However, the RSS, denoted by $S(t)$ in Equation (6), could be formulated based on the Friis transmission formula [39]:

$$S(t) = S_{Tx} G_t G_r \frac{\lambda}{4\pi d^2} \cdot R_{vessel} \cdot A_{skin} \cdot A_{blood}(t), \tag{9}$$

where S_{Tx} denotes the transmitting signal's strength, G_t and G_r are the transmitting and receiving antenna's gain, d indicates the distance between the artery and radar, R_{vessel} denotes the reflection coefficient at the bottom surface of blood vessels, and A_{skin} and A_{blood} are absorption coefficients of skin and blood flow. R_{vessel} and A_{skin} could both be considered constant in our experiments. Hence, the change of $S(t)$ (denoted by $\Delta S(t)$) is only related to diameter changes of blood vessels $\Delta L(t)$, which could be derived by substituting Equation (8) into Equation (9) to obtain the following formula:

$$\Delta S(t) \propto e^{-\Delta L(t)}, \tag{10}$$

which implies that the RSS variation of mmWave signals reflected by the wrist is negatively correlated with the wrist artery's diameter change.

Therefore, we could capture the wrist arterial pulse leveraging RSS variation, which provides effective hints for blood volume change. A showcase is illustrated in Figure 5(b). When the blood flow volume is dilated to its maximum at t_1, RSS drops to its lowest point. However, RSS increases to the maximum at t_2 owing to the decrease of blood volume, and it drops again before t_3 due to the reflected blood flow.

2.3 From Wrist Artery Sensing to BP Estimation

Although the ability to sense wrist arteries via RSS has been validated, we note that BP estimation using contactless mmWave is still challenging. When using contactless mmWave to sense the wrist from a distance (e.g., 12 cm), the captured mmWave signal contains not only the reflection of the tiny arteries but also reflection from other much larger parts of the wrist, like muscles and veins. Our purpose is to get the reflection of arteries, in which case the reflection of other parts could be considered interference. So we design ACC-maximizing beam-forming to focus on the reflection of arteries using the inherent strong periodicity of the arterial pulse. However, there are still two main obstacles preventing airBP from obtaining high-quality and fine-grained pulse waveforms (i) signals corresponding to the minor arterial activity (only 0.03 mm) are rather weak from a distance (e.g., 12 cm), and (ii) the hindrance of skin and muscles that wraps the arterial vessels makes the effective signal harder to clearly sense in a contactless way.

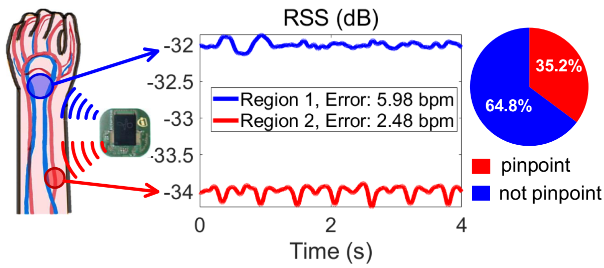

Fig. 6. RSS-maximizing beam-forming fails to pinpoint the wrist artery, which should be the red region instead of blue. The pinpoint ratio of the best region candidate is only 35.2%.

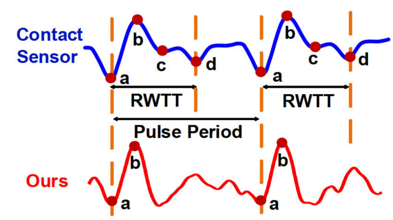

Fig. 7. Compared to the contact sensor, mmWave generates an ambiguous pulse waveform, of which it is hard to extract BP-related features (i.e. RWTT).

Difficulty in Pinpointing the Tiny Wrist Artery. To isolate the reflection from the 3-mm tiny-sized artery, a very careful and tedious searching process is required. One natural solution is digital beam-forming. In particular, one can leverage the antenna phased array on the mmWave radar [29] to separate receiving signals reflected from different zones, and pick the one with the largest RSS, and deem it as the one from the target zone (i.e., the wrist artery). Unfortunately, we find that such *RSS-maximizing beam-forming* does not fit. In particular, we compare the algorithm's output against the ground truth derived by a patient monitor [40] and give the results in Figure 6. We observe that only 35.2% of arteries are successfully pinpointed by RSS-maximizing beam-forming, whereas 64.8% of them come from other tissues (e.g., arm muscles). Moreover, such pinpointing failures translate to 43.3% larger BP estimation error, as shown in Section 7.3. Figure 6 shows that the zone with the maximum RSS (the blue spot) leads to an inaccurate heart rate estimation (i.e., a larger MAE of 5.98 bpm), whereas the real target (the red spot) with lower RSS is missed. The results motivate us to design more efficient beam-forming by exploiting the domain knowledge of the wrist artery's cardiac motion in Section 4.

Difficulty in Precisely Extracting BP-Related Features. Once the pulse waveform has been derived, the next step is to extract BP-related features (i.e., RWTT) from the pulse, then BP can be computed using such features. However, it requires accurately capturing the fine-grained details of the minor arterial activity, which is hard to sense for contactless mmWave from a certain distance. As shown by the blue line in Figure 7, the existing contact sensor based method [21] generates high-quality pulse waves using the quadratic differentiation of PPG signals, where RWTT could be easily estimated from specific data points (point a, b, c, d). Yet, for pulse waves generated by contactless mmWave (the red line in Figure 7), such specific data points (i.e., points c and d) are hard to accurately obtain. This is mainly due to the challenging task for contactless mmWave to sense the minor artery dilation caused by a small amount of reflected blood flow from a distance (e.g., 12 cm). Consequently, we propose our new designs to estimate BP using such signals of a low signal-to-noise ratio, as given in Section 5.

3 SYSTEM DESIGN

Motivated by the preceding measurement observations, we make a different design choice for airBP—that is, we directly estimate BP from the arterial pulse waveform, by skipping the explicit RWTT calculation as done in previous works [21]. We make such a choice due to the following considerations. First, RWTT is hard to explicitly compute from the ambiguous pulse waveform generated by "in-air" mmWave. Second, although one may try machine learning approaches to get RWTT implicitly, the ground-truth RWTT is necessary for training. However, the measurement

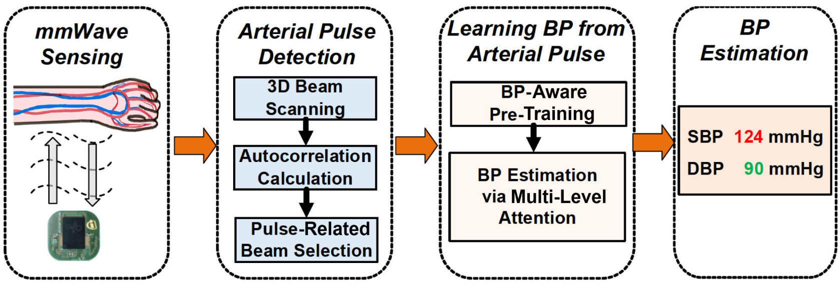

Fig. 8. System overview of airBP.

of actual RWTT ground truth is injurious and impractical, which requires intrusively implanting medical micro-accelerometers inside the artery [41].

We present airBP's system overview in Figure 8, which is composed of three steps. In particular, airBP first transmits mmWave signals toward a user's wrist and then captures the reflected signal bounced off the wrist. Then airBP analyzes the signal to detect the tiny wrist artery motion and obtain the arterial pulse. Finally, airBP estimates BP from the arterial pulse waveform with a customized deep learning design. To accomplish the whole process, the key design modules are as follows.

Arterial Pulse Detection. As shown in Section 2.3, the traditional RSS-maximizing beam-forming could not pinpoint arteries effectively. Thus, airBP instead focuses on the specific sub-zone where the recovered pulse exhibits the maximum ACC, leveraging the strong inherent periodicity of the arterial pulse. In particular, airBP first exploits multiple receiving antennas on the mmWave radar to obtain reflected mmWave signal portion from each sub-zone. Then airBP calculates the ACC of each portion and denotes the sub-zone with the maximum ACC as the target (i.e., the wrist artery). Finally, airBP recovers the pulse waveform from the RSS variation corresponding to the target zone.

Learning BP from the Arterial Pulse. We first design and *pre-train* a learning model with stronger supervision—that is, a long-series representation of BP-related data points instead of a single numeric BP value. The pre-training helps airBP efficiently generate discriminative BP-related features. Moreover, we customize a convolutional transformer learning model with the multi-level attention mechanism, which exploits the pulse's own characteristics (different from information-redundant images or speech)—that is, inherent periodicity of BP-related features and contextual similarity among pulse cycles—so as to get accurate BP estimation.

In the following, we proceed to detail these two modules.

4 ARTERIAL PULSE DETECTION

We first show that an arterial pulse exhibits strong periodicity, which can be quantized by a metric called *ACC*. Then we describe how to leverage this fact for wrist artery detection and arterial pulse recovery.

4.1 Strong Periodicity of the Arterial Pulse

We note that a time series of the arterial pulse has a very strong periodicity, since the arterial pulse roots from periodical cardiac pulsation. Specifically, we employ ACC [42] to measure the periodicity of time-series RSS data, denoted by $S(t)$ in Equation (6). ACC quantifies the correlation degree between $S(t)$ and $S(t+l)$, which are the same time-series data starting from t and $t+l$ with the lagged time l. The calculation could be formulated as follows:

$$ACC(l) = \sum_{t=0}^{N_t-1-l} \frac{S(t) \cdot S(t+l)}{N_t - l}, \tag{11}$$

where N_t is the length of the whole time-series data. Larger ACC values indicate stronger periodicity over time. We further analyze the 287 arterial sensing results mentioned in Section 2.3 and separately plot their estimated pulse rate error and ACC on the vertical and horizontal axis of Figure 10(a), which is shown later. In that figure, we divide the data into two groups: low-ACC and high-ACC data, with an ACC threshold of 0.4. We find that 32.4% of the data has low ACC, and the **Limit of Agreement (LOA)** [43] (defined as ME ± 1.96 ×STD) of low-ACC data is [−21.85, 24.36], whereas that of high-ACC data is [−8.90, 9.99] bpm. We note that more than 95% of data lies within the LOA, which indicates that the more periodic data has a higher agreement with the arterial pulse rate. This inspires us to detect the arterial pulse by leveraging the periodicity characteristics: instead of the signal portion with the maximum RSS, the one with the maximum ACC more likely corresponds to the reflection of wrist arteries.

4.2 ACC-Maximizing Beam-Forming

Our arterial pulse detection process is illustrated in Figure 8. We first transmit the mmWave toward the wrist and scan the signals reflected from the 3-dimensional space, segmented into sub-zones by different azimuth/elevation angles, and ranges. Then we separately calculate the ACC of each sub-zone and select the one with the highest ACC, and deem it as the target artery.

The detailed process of ACC-maximizing beam-forming is present in Figure 9. To scan the 3-dimensional space, we leverage the multiple receiving antennas (Rx) of mmWave radar. Given an FMCW mmWave radar with the Rx array of $M_\theta \times M_\beta$, where M_θ is the number of Rx in the azimuth direction and M_β is the number of Rx in the elevation direction, the output of the radar in one chirp, denoted by $x[\tau_n]$ in Section 2.1, is a matrix with the dimensions of $M_\theta \times M_\beta \times N_\tau$, where N_τ is the number of samples in one chirp. In the following, we define the output of the Rx of the ith in azimuth direction and the jth in elevation direction sampled at the discrete time τ_n within one chirp as $x_{i,j,n}$, where $i = 0, 1, \ldots, M_\theta - 1$, $j = 0, 1, \ldots, M_\beta - 1$, and $n = 0, 1, \ldots, N_\tau - 1$.

First, as illustrated by Step 1, Step 2, and Step 3 in Figure 9, we separately apply the Discrete Fourier Transform $\mathscr{F}$ on each of the three dimensions, which could be formulated as follows:

$$X_{p,q,k} = \mathscr{F}_j[\mathscr{F}_i[\mathscr{F}_n[x_{i,j,n}]]] = \sum_{j=0}^{M_\beta-1} \sum_{i=0}^{M_\theta-1} \sum_{n=0}^{N_\tau-1} x_{i,j,n} \cdot e^{-j2\pi(\frac{kn}{N_\tau} + \frac{pi}{M_\theta} + \frac{qj}{M_\beta})}, \tag{12}$$

where k is the range bin index corresponding to the radar-target range d within $[\frac{kc}{2B}, \frac{(k+1)c}{2B}]$ (refer to Section 2.1 for details), whereas p and q are the angle bin index [45] corresponding to the azimuth angle of arrival θ and elevation angle of arrival β. As a result, $X_{p,q,k}$ could be considered as $X_{\theta,\beta,d}$, which are reflected signals from 3-dimensional space (different ranges d, azimuth angles θ, and elevation β angles).

Then as shown by Step 4 of Figure 9, we extract the RSS variation over time and further calculate the corresponding ACC value to measure the periodicity. Specifically, we set the radar to transmit one chirp per frame and record a series of chirps to obtain the variance of $X_{\theta,\beta,d}$ over time t, as denoted by $X_{\theta,\beta,d}(t)$. Recalling Equation (6), $X_{\theta,\beta,d}(t)$ could be formulated as follows:

$$X_{\theta,\beta,d}(t) = S_{\theta,\beta,d}(t) \cdot e^{j\phi_{\theta,\beta,d}(t)}, \tag{13}$$

from which we extract the RSS variance over time $S_{\theta,\beta,d}(t)$. After that, we calculate the $ACC_{\theta,\beta,d}(l)$ according to Equation (11) and extract the maximum value across different lagged time l, which is

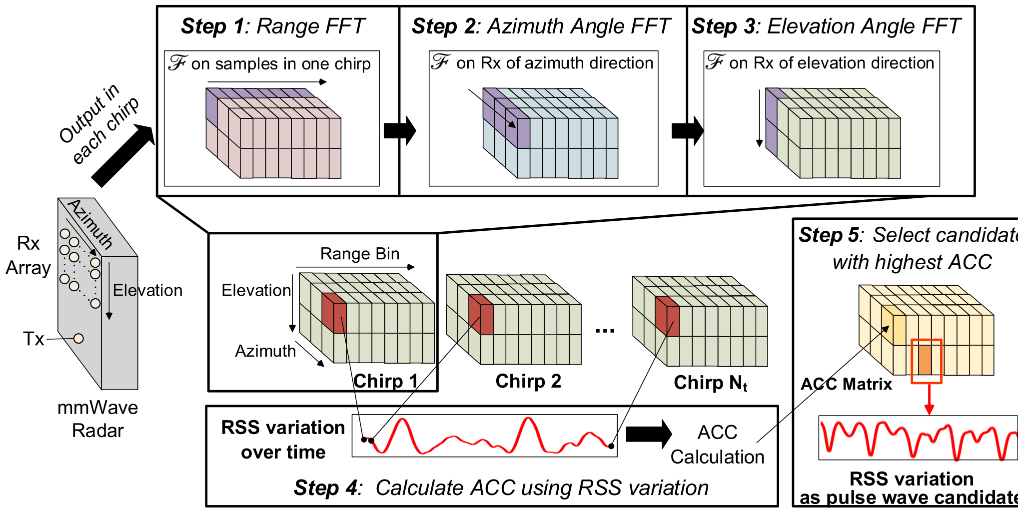

Fig. 9. The process of ACC-maximizing beam-forming. We first obtain signals reflected from 3-dimensional space (ranges, azimuth angles, and elevation angles) by applying the Discrete Fourier Transform $\mathscr{F}$ on each chirp's output along each of the three dimensions (samples per chirp/azimuth direction/elevation direction). Then we record the RSS variation over time (by a series of chirps) for reflection from each 3-dimensional sub-space and calculate ACC to measure the periodicity. Finally, the sub-space with the maximum ACC values (illustrated by the orange color marked by the red box) is selected as the location of artery, from which we obtain the RSS variation with strong periodicity as the pulse wave candidate.

achieved when l is equal to the pulse period. The calculation could be formulated as follows:

$$\overline{ACC}_{\theta,\beta,d} = \max ACC_{\theta,\beta,d}(l). \tag{14}$$

Finally, as shown by Step 5 of Figure 9, $\overline{ACC}_{\theta,\beta,d}$ values of different 3-dimensional sub-space (θ, β, d) make up an ACC matrix (illustrated by the yellow color matrix in Step 5 of Figure 9), which indicates the periodicity of RSS variation over time in each sub-space. Note that the pulse activity should have the strongest periodicity, so we scan $\overline{ACC}_{\theta,\beta,d}$ values across different sub-space to get the one with the maximum values, whose coordinates are denoted as $(\hat{\theta}, \hat{\beta}, \hat{d})$. This corresponds to the sub-space where the wrist artery lies. The process can be expressed as follows:

$$(\hat{\theta}, \hat{\beta}, \hat{d}) = \arg\max_{\theta,\beta,d} \overline{ACC}_{\theta,\beta,d}. \tag{15}$$

Then we extract RSS variation over time from the selected sub-space, which is denoted as $S_{\hat{\theta},\hat{\beta},\hat{d}}(t)$, and adopt variational mode decomposition [46] for de-noising so as to obtain the arterial pulse wave z. This is further taken as the input of the learning-based BP estimator to estimate BP.

We note that although the wrist is shown in different range bins, our ACC-maximizing beam-forming method only chooses the range bin that has the maximum ACC as the best candidate to sense the arterial pulse, leveraging the inherent strong periodicity of arterial pulse activity. Theoretically, radial arteries in the wrist across multiple range bins could all be considered good candidates. However, different parts of them are buried at different depths under the skin. Our ACC-maximizing beam-forming method will automatically choose the range bin where radial arteries have the smallest depth under the skin, since the hindrances of skin and muscles are less than other parts and could provide clear arterial pulse activity with a larger ACC.

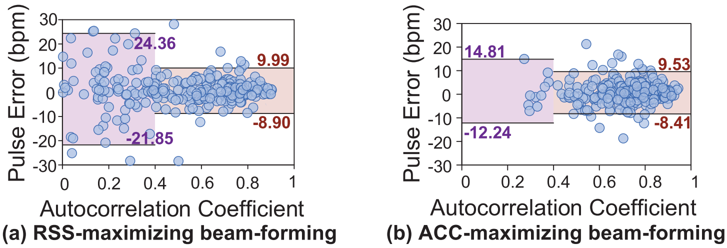

Fig. 10. Pulse rate estimation of RSS-maximizing beam-forming (a) and our ACC-maximizing beam-forming method (b). Note that the artery's misalign rate (error >10 bpm) drops from 12.5% to 4.9% by applying our method.

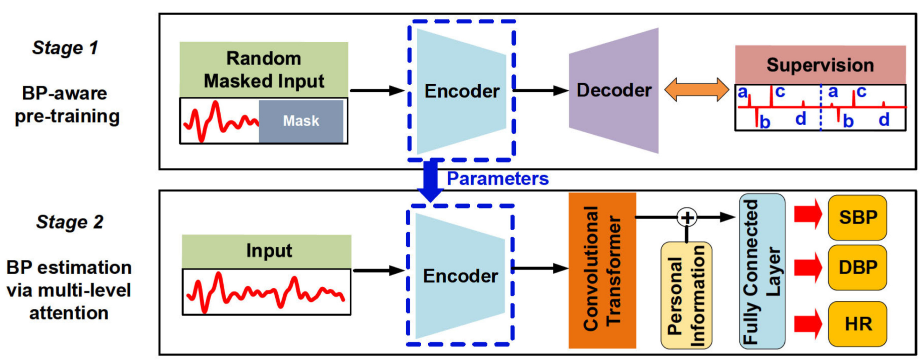

Fig. 11. Overview of airBP's two-stage learning mechanism for BP estimation.

Benefits of Accurate Beam-Forming. We compare the estimated pulse rate, by RSS-maximizing and ACC-maximizing approaches, respectively, against the ground truth recorded by an FDA-approved device [32]. From the results in Figure 10, we observe the following. RSS-maximizing leads to many inaccurate detections with very large pulse errors (i.e., up to 24.26 bpm in the purple box of Figure 10(a)), which indicates that it focuses on wrong tissues. However, ACC-maximizing provides more accurate artery sensing results (i.e., improved ACC as well as lower errors on the estimated pulse rate). We will further validate its gain on BP estimation performance in Section 7.3.

5 LEARNING-BASED BP ESTIMATION

In practice, we find that directly reusing the learning-based methods in previous contact-based BP studies (e.g., BLSTM [21, 47]) leads to overfitting and thus poor BP estimation results. The main reason is the challenging task to discriminate sparse BP-related data points from the ambiguous and noisy mmWave-generated pulse waveform. To handle the issue, we present a custom-designed learning-based BP estimator as shown in Figure 11, which consists of two stages: BP-aware pre-training and BP estimation via multi-level attention.

BP-Aware Pre-Training. To enhance the awareness of sparse BP-related data points, we first pre-train an auto-encoding model with stronger supervision. The auto-encoding model is composed of an encoder and a decoder, which has achieved remarkable success in speech and image processing [48, 49], due to its powerful ability of feature representation. It first automatically encodes a long data series input into latent features, then reconstructs the long-series representation using

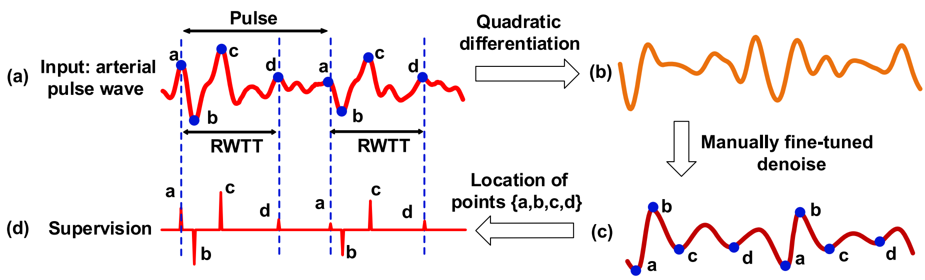

Fig. 12. Generation of the long-series supervision used for pre-training. Taking the input arterial pulse wave as shown in (a), we first perform quadratic differentiation to get the acceleration of blood vessel extension shown in (b), and then apply a manually fine-tuned bandpass FIR filter to eliminate noise and get location of BP-related points a,b,c,d as shown in (c). Finally, we set the other data points of pulse wave to 0 except the preceding specific points to obtain supervision used in pre-training as shown in (d).

the extracted features. In our case, such a mechanism is adopted with customized input and supervision to realize its potential in BP estimation. First, we randomly mask 50% of inputs in pre-training to enhance the model's feature extraction ability, which enforces the model to learn latent representations from only a part of long time-series data. Second, we design a "signal skeleton" as stronger supervision instead of using the original samples (commonly used in the pre-training stage of speech/image processing [48, 50]). In this way, airBP enforces discriminating BP features from the other irrelevant features and translates BP-related data points into compact latent features.

BP Estimation via Multi-Level Attention. For the sake of BP estimation, we find that the widely used BLSTM [21, 47] fails to capture long-term dependencies [51]. In airBP, we design a multi-level attention mechanism by combining global attention and local contextual attention. The former enables utilization of BP-related features from the whole series regardless of long separation among key points, whereas the contextual attention takes advantage of the context information (i.e., periodically repeating BP-related features) so as to achieve accurate BP estimation.

We combine these two stages by reusing the encoder, which transfers its pre-training experience into effective BP estimation. Meanwhile, we perform auto-calibration using personal information in the second stage, so as to deliver user-independent BP monitoring.

5.1 BP-Aware Pre-Training

Masked Input Generation. As shown in Figure 11, the pre-training stage takes the arterial pulse waveform z and randomly masks 50% successive data points of z to zero, generating a marked input $\bar{z}$. We note that the masking ratio is set to 50% since it achieves the best BP estimation performance in our experiments when it ranges from 15% to 75% following the existing experience [48, 50]. We attribute this result to the fact that a 50% masking ratio can generate the most diverse input, so as to avoid overfitting.

Generation of the Strong Long-Series Supervision. The generation of supervision O_g used in pre-training (i.e., the signal skeleton) is shown in Figure 12. In particular, we set most data points of the input arterial pulse waveform z (see Figure 12(a)) as 0 except those related to BP and pulse period, denoted by $\{a, b, c, d\}$. The location of these points is calculated following the traditional RWTT computation [21] First, we calculate the quadratic differentiation of z, which is illustrated in Figure 12(b). The quadratic differentiation of the pulse wave represents the acceleration of blood vessel extension, which has more high-frequency details related to BP [21, 52, 53]. Second, then we get the location of BP-related points illustrated in Figure 12(c) as follows. We identify point b as the maximum point in each pulse period, and points a and c as the local minimum points before

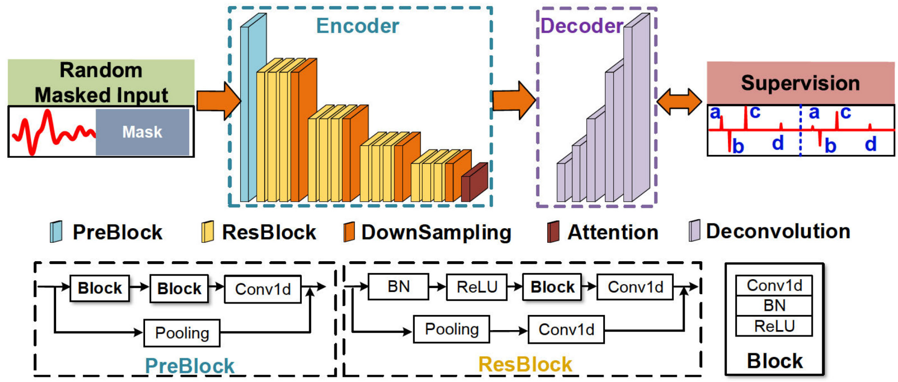

Fig. 13. Architecture of the encoder and decoder.

and after b. To obtain the location of point d, which is identified by the next local minimum points after c, we apply a manually fine-tuned bandpass FIR filter to eliminate noise while preserving this RWTT-related point. Specifically, we carefully fine-tune the band parameters of the bandpass FIR filter according to the heart rate and BP of each subject. We first set the lower band to 0.4 Hz and the upper band to six times the frequency of heart rate. Then we gradually reduce the upper band to four times the frequency of heart rate with a step of 0.2 Hz while checking the correlation between the calculated $\frac{1}{RWTT}$ and recorded SBP. Then we choose the parameters with the highest correlation (shorter RWTT corresponds to higher BP of the same subject) and obtain the location of point d. Third, taking the location of $\{a, b, c, d\}$, we could generate supervision O_g used in pre-training (i.e., the signal skeleton) by setting the other data points of pulse wave to 0 as shown in Figure 12(d). Note that such supervision is only generated for the training set in the pre-training and is not required for the test subject's online BP estimation. Hence, the supervision generation process will not hinder the system's practical usage.

Auto-Encoding BP-Related Features. Taking the masked data $\bar{z}$ as input, we further extract BP-related features with a custom-designed encoder q_ψ, which discriminates BP features from the other irrelevant features and further generates latent features h. The extraction can be formulated as $h = q_\psi(\bar{z})$, $h \in \mathbb{R}^{m \times d_h}$, where m and d_h denote the dimension of features and time-series length. We implement this using a custom-designed encoder, as shown in Figure 13, which consists of 1 PreBlock, 16 ResBlocks, and 1 attention layer. Every 4 ResBlocks sub-sample the input by a factor of 2 to reduce the scale of redundant features. In addition, we add (i) a residual connection in each ResBlock to reduce the gradient disappearance for deep learning models [54, 55] and (ii) an attention block to focus on the most informative BP features among the long series. Specifically, it applies scaled dot-product attention [56] formulated by

$$h = Softmax(WA(Y^T))Y, \tag{16}$$

where Y and W are the attention block's inputs and learnable parameters, respectively. A denotes the activation function $tanh$, which could be formulated by

$$A(y) = \frac{e^y - e^{-y}}{e^y + e^{-y}}. \tag{17}$$

Training with Strong Supervision. After obtaining latent features h, we further translate them into long-series representation O_p, which has the same size as supervision O_g. The translation can be formulated as $O_p = p_\Psi(h)$, which is implemented with a light decoder p_Ψ composed of five

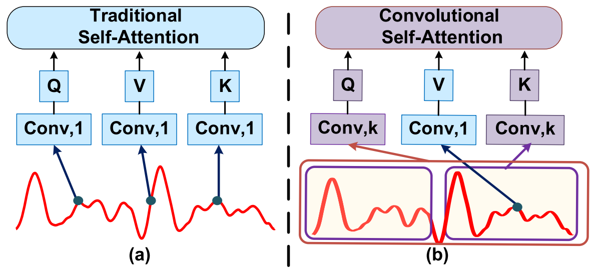

Fig. 14. airBP's convolutional transformer enhances context awareness. (a) The traditional transformer only uses convolution of kernel size 1 to extract features, which focuses on the data with similar values (blue points). (b) Our convolutional transformer leverages convolution with a larger kernel size k to be aware of BP-related features (purple boxes) that appear periodically and thus enables accurate BP estimation.

stacked deconvolutional layers as shown in Figure 13. The generated O_p is supposed to contain data points related to BP and pulse period. Thus, we train the learnable parameters in encoder q_ψ and decoder p_Ψ with the following training loss:

$$Loss = \sum (O_p - O_g)^2. \tag{18}$$

Now we obtain a pre-trained encoder, and we further transfer this ability into BP estimation described in the following.

5.2 BP Estimation Via Multi-Level Attention

As illustrated by the second stage in Figure 11, we design a hybrid model to enable accurate estimation leveraging the multi-level attention mechanism. Specifically, the model is composed of an encoder with pre-trained parameters, a convolutional transformer, and a fully connected layer, which harmoniously combines personal information and BP-related features.

Transfer the BP-Awareness into the Estimator. As shown in Figure 11, the BP estimator has an encoder with the same structure as the pre-trained one q_ψ. We first transfer the learnable parameters ψ of the pre-trained encoder q_ψ into that of BP estimator, so as to transfer the awareness of sparse BP-related data points. While training the estimator, we freeze the parameters of the encoder and only train parameters belonging to the other module.

Design of the Convolutional Transformer. The convolutional transformer is made up of three convolutional self-attention layers. We present a showcase in Figure 14(b), which enables learning with a global attention mechanism and can be formulated as follows:

$$O_h = Softmax\left(\frac{QK^T}{\sqrt{d_h}}\right)V, \tag{19}$$

where O_h denotes the output features and Q, K, V are generated from input features h using learnable parameters in convolutional layers. They all have the same size of h and are utilized to extract global similarity by employing matrix multiplication. In addition, convolutional self-attention layers enhance the awareness of local contextual information with larger kernel size k of convolutional layers.

For a better understanding, we show the difference between the traditional and convolutional transformer in Figure 14. The traditional transformer only uses convolution of kernel size 1 to generate features Q, K, V, which focuses on the data with similar values (blue points). In contrast, the convolutional transformer leverages convolution with a larger kernel size k to be fully aware

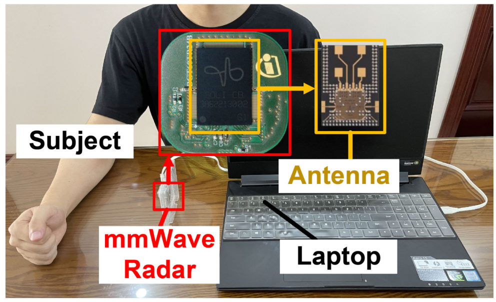

Fig. 15. Prototype of airBP.

of BP-related features (purple boxes) that appear periodically, and thus enables accurate BP estimation. Finally, we apply max-pooling on features O_h extracted by the convolutional transformer and obtain the BP-related features $O_{BP} \in \mathbb{R}^m$ for BP estimation.

Auto-Calibration for User Independence. To provide user-independent BP monitoring, we further collect personal information PI = {gender, age, height, weight} of each subject to perform auto-calibration. Specifically, we perform normalization on PI using its mean and STD and further concatenate it with features extracted by convolutional transformer O_{BP} to build the input tensor O_{input}. Then, as shown in Figure 11, the tensor O_{input} is fed into a fully connected layer to generate BP and heart rate estimation: $O_{pre} = \{SBP_{pre}, DBP_{pre}, HR_{pre}\}$. We employ the same loss as described in Equation (18) to train the predictor under the supervision of the ground truth of SBP, DBP, and heart rate values.

6 IMPLEMENTATION

Hardware. We build airBP using a coin-size COTS mmWave radar BGT60TR24, as shown in Figure 15, which works with FMCW at a frequency from 57 to 63 GHz [30]. The power consumption of the radar is only 300 mW, which could be deployed on mobile low-power devices [44], such as smartwatches and smartphones. The radar is connected to a laptop, conducting data collection and running airBP. The machine learning model is trained on a cloud server with a Tesla V100 GPU and deployed on the laptop to perform BP inference, where airBP runs in real time. We note that the laptop could also be replaced by other mobile devices, such as a smartphone or smartwatch, for applications in various scenarios.

Radar Settings. We utilize one transmitting antenna and four receiving antennas to sense the artery. The ADC sample rate is set to 2 MHz, and the chirp duration time is 32 μs. During each chirp, the FMCW radar takes 256 samples, which corresponds to the range resolution of 2.5 cm. The number of chirps per frame is set to 7. The overall sample rate for pulse wave sensing is 210 Hz, which is sufficient for pulse recovery.

Details of Signal Processing. The FMCW radar outputs the intermediate frequency signal in each chirp (as defined by $x[\tau_n]$) in Equation (3), then we record it from four receiving antennas for about 35 seconds. After that, we implement airBP's signal processing algorithm using Python on the laptop.

We first perform ACC-maximizing beam-forming to extract the reflection from arteries. The process is as illustrated in Figure 9. We implement range-FFT and angle-FFT using Discrete Fourier Transform. Range-FFT has no padding. Angle-FFT is performed on each pair of adjacent azimuthal/elevational antennas, generating six angle bins with padding. In practice, to reduce computation costs, after performing range-FFT, we first select range bins within 50 cm (the 1st to 20th bin) from each of the four receiving antennas to calculate ACC values and select the one with the maximum ACC value to determine the range of artery $\hat{d}$. Then we separately scan the six azimuthal angle bins and six elevational angle bins at the range bin of $\hat{d}$ to determine the azimuthal and elevational arrival angle of the artery's reflection. In summary, there are 92 candidates to scan (80 zones of different range bins and 12 potential angle candidates), and we select the one with the maximum ACC value to get the reflection of arteries.

After that, we perform data de-noising on the RSS variation of artery reflection over 35 seconds. Specifically, we perform variational mode decomposition [46], which is an effective method for signal decomposition based on their respective center frequencies. We divide the receiving signals into nine components and select components of the second and third lowest frequency to combine and generate the output arterial pulse wave. In this way, the low-frequency and high-frequency noise are both filtered. We note that the signal processing of variational mode decomposition employs constant parameters and does not require fine-tuning for different data. So it is applied as the input signal de-noising method for both training and test datasets before feeding into neural networks. Although the manually fine-tuned bandpass FIR filter could achieve the best de-noise performance, it requires extra information of heart rate and BP, which is unknown for test subjects. So the fine-tuned de-noise method is only used when generating the signal skeleton supervision for the data of the training set.

Details of Learning Model Design. The learning model is built with PyTorch [57]. In the encoder, the kernel size of all 1-dimensional convolutional layers is 3, and we implement pooling layers as max-pooling with the kernel of 3. Output channels of all convolutional layers in PreBlock are 32, whereas ResBlocks have the same output channels before each downsampling and increase by a factor of 2 after that, which are {32, 64, 128, 256}. The kernel size of all deconvolutional layers in the decoder is 2, whereas the output channels are {128, 64, 32, 16, 8, 1}. In the convolutional transformer, we employ three multi-head convolutional self-attention layers, each having eight heads, and the kernel size k is set to 5. Finally, it generates 256 features to concatenate with the 4-dimensional personal information.

Details of Model Training. The model training contains two stages. In each stage, we set the batch size as 20 and train the model for 300 epochs using the SGD optimizer with an initial learning rate of 0.001. In the first stage (i.e., pre-training), the data length of each pulse wave is set as 6,300, where we randomly select consistent 6,300 samples from every 35-second arterial pulse wave (which usually has more than 7,000 samples). After the pre-training, we transfer the trained parameters of the encoder into the other encoder used for the following BP estimation. In the second stage, we freeze the transferred trainable parameters of the encoder and only train parameters of the convolutional transformer and fully connected layer. After the training of two stages using data from the training set, we obtain the model for BP estimation that could be directly used for the test subject without extra pre-training.

7 EVALUATION

7.1 Experimental Setup

Subjects. We obtained IRB approval to conduct experiments and collect data from 41 diverse subjects with the demographic in Table 2. To enrich the diversity, we deliberately recruit 8 volunteers with rather low BP (SBP of 80–100 mmHg) and 20 with high BP (SBP of 130–160 mmHg). We employ the **Body Mass Index (BMI)** [58] to measure the body fat index, according to which 12 are overweight (BMI: 25–30 kg/m^2) and 4 are obese (BMI: ≥30 kg/m^2). To investigate the impact of skin tone, we categorize subjects' skin tones according to the Fitzpatrick classification scale [59]. They are divided into three classes: lighter, common, and darker, corresponding to types III, IV, and V of the Fitzpatrick scale, respectively. We note that all participants are Asian, and we plan to further test airBP on African and White people in the future.

Table 2. Demographic Data of Participants

Demographics		No.	Demographics		No.
Gender	Woman	11	Age (years)	≤40	33
	Man	30		>40	8
BMI (kg/m^2)	≤25	25	Skin Tone	Type III	11
	25–30	12		Type IV	26
	≥30	4		Type V	4
SBP (mmHg)	80–100	8	DBP (mmHg)	50–70	14
	100–130	20		70–90	24
	130–160	13		90–100	3

Data Collection. As shown in Figure 15, each subject is asked to sit in a chair and put the forearm in front of the mmWave radar at a distance of 12 cm. Subjects adjust the height of the chair to keep the wrist at heart level and keep the arm still for 35 seconds during each round of measurement with mmWave. Then we immediately measure the ground-truth SBP, DBP, and heart rate using an FDA-approved wrist-cuff Omron BP measurement device [32]. The ground truth is measured on the same wrist as airBP. Additionally, it is measured twice at intervals of 2 minutes and averaged to reduce errors, following recommendations of the American Heart Association [60]. We note that one natural idea to obtain the ground-truth reference is measuring the reference BP on the other wrist or arm while measuring BP using airBP simultaneously. However, there are inter-arm BP differences between the right and left arms [61], which means that it could not be considered as accurate reference BP values. So we measure the reference BP on the same wrist. BP values usually do not vary when sitting still. Medical literature shows that two BP measurements at 5-minute intervals of sitting rest are consistent [62]. So the reference BP is effective during each measurement.

Before the next measurement using mmWave and Omron, each subject is asked to perform 2 minutes of free movement, such as walking and bike riding, to obtain various BP values following existing literature [19]. The whole measurement for each participant takes about 1 hour, during which participants repeat the preceding process six to eight times. Unless otherwise noted, experiments are conducted in a lab at 20°C temperature with common ambient light conditions. We totally collect 287 data from 41 subjects. Each data has a duration of 35 seconds and contains more than 35 pulse periods. In summary, more than 10,000 pulse periods are contained in the dataset. To validate the robustness of airBP, we further collect extra data from eight subjects under different environments, distances, wrist angles, and light conditions. These data are only for testing—that is, using the model trained on lab data.

Evaluation Metrics. We employ the widely adopted metrics in BP studies to evaluate airBP performance: ME, STD error, and **Pearson Correlation Coefficient (PCC)**. In addition, we employ a stricter metric of MAE, which emphasizes the absolute error in each measurement rather than the ME (representing the average over multiple measurements).

7.2 System-Level Evaluation

We evaluate airBP from two aspects: (i) overall performance comparison with state-of-the-art eBP [16] and Crisp-BP [21], and (ii) performance over multiple impact factors (i.e., different kinds of environments, target distances, wrist angle, and ambient light conditions).

Overall Performance. We first perform the leave-one-subject-out validation [21] (i.e., training with data of 40 subjects and testing on the remaining 1 subject), which ensures that airBP is

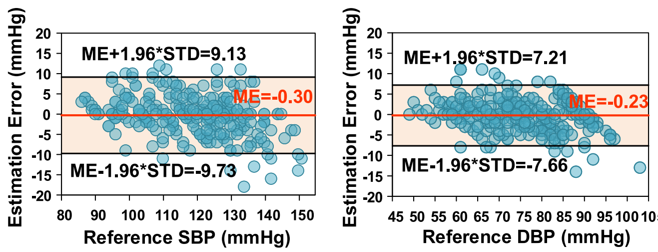

Fig. 16. Bland-Altman diagram of airBP's estimation results in SBP and DBP.

Table 3. Comparison in Terms of the BHS Standard

		Cumulative Error Percentage		
		≤5 mmHg	≤10 mmHg	≤15 mmHg
BHS [63]	Grade C	40%	65%	85%
	Grade B	50%	75%	90%
	Grade A	60%	85%	95%
SBP Estimation	Crisp-BP [21]	65.66%	87.17%	96.23%
	airBP	**76.66%**	**95.47%**	**98.95%**
DBP Estimation	Crisp-BP [21]	76.73%	91.43%	97.96%
	airBP	**86.41%**	**97.56%**	**99.65%**

evaluated on unseen independent subjects. Results are as follows. First, in Figure 16, we plot results of airBP in a Bland-Altman diagram, which charts the estimation error and reference BP on the vertical and horizontal axis, respectively. ME and STD of airBP's estimation are −0.30 and 4.80 for SBP and −0.23 and 3.79 for DBP, which meets the standard requirements of the FDA's AAMI [31] for a medical BP device (i.e., ME ≤ 5, STD ≤ 8). Meanwhile, airBP shows quite acceptable results with the LOA [43], defined as ME±1.96 ×STD. The LOA of SBP and DBP are [−9.73, 9.13] mmHg and [−7.66, 7.21] mmHg, respectively. We find that more than 95% of data lies between the LOA. Second, as shown in Table 3, airBP also meets the requirements of the BHS (Britain Hypertension Society) [63] standard and gets Grade A[1] for both SBP and DBP estimation. Specifically, 76.66% of SBP and 86.41% of DBP estimation lies within 5-mmHg errors. This confirms that airBP can ensure accurate and user-independent BP estimation across various users.

Comparison against the State of the Art. We compare our contactless BP monitoring system airBP to two state-of-the-art contact-based systems: eBP [16], an in-ear system that captures BP with a tiny pump in the ear, and Crisp-BP [21], the most recent wearable BP monitor with a PPG sensor in a wrist-worn device.

First, we list the reported results of eBP and Crisp-BP against our results in Table 4. We observe that airBP achieves higher accuracy compared to the existing best results: that of Crisp-BP, with only 18% ME and 66% STD for SBP, and 27% ME and 58% STD for DBP. Our airBP also achieves remarkable improvements on PCC, from 0.88 and 0.82 to 0.94 in terms of SBP and DBP. Meanwhile, as shown in Table 3, airBP outperforms Crisp-BP with more accurate estimation—for example, 76.66% of SBP and 86.41% of DBP prediction errors are within 5 mmHg, which is more than 10% percent higher compared to Crisp-BP.

[1]More than 60% estimation errors are within 5 mmHg, more than 85% estimation errors are within 10 mmHg, and more than 95% are within 15 mmHg.

Table 4. Comparison with State-of-the-Art BP Monitors

Name	eBP [16]	Crisp-BP [21]	*airBP*
Type	Compress	Contact	*Contactless*
Subject	35	35	*41*
ME: SBP/DBP	1.8 / -3.1	1.67 / 0.86	*−0.30/−0.23*
STD: SBP/DBP	7.2 / 7.9	7.31 / 6.55	*4.80/3.79*
PCC: SBP/DBP	0.81 / 0.76	0.88 / 0.82	*0.94/0.94*

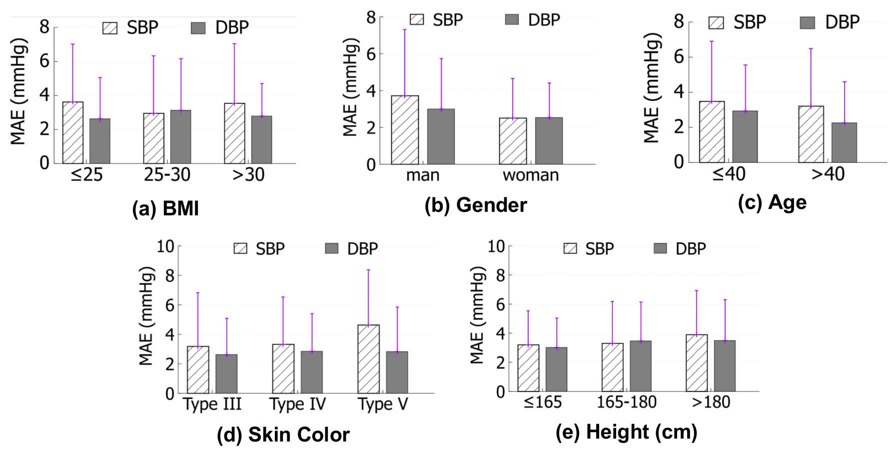

Fig. 17. BP estimation error of airBP on different kinds of subjects.

Second, we implement Crisp-BP [21] and test it on our mmWave signal's dataset, even though it is designed for the contact-based PPG signal. In other words, Crisp-BP first extracts RWTT from the pulse waveform and then utilizes a learning-based model to estimate BP using RWTT. We observe that the ME of Crisp-BP's estimation is −6.09 mmHg, which is more than 20 times that of airBP (i.e., −0.30 mmHg). Meanwhile, its LOA is [−23.99, 11.81] mmHg, which is 1.9 times that of airBP (i.e., [−9.73, 9.13] mmHg). The reason lies in Crisp-BP's inaccurate RWTT estimation when applied to the noisy arterial pulse wave generated by mmWave in Crisp-BP, as validated in Figure 7.

Impact of Subject Diversity. We present airBP's MAE across different kinds (BMI, gender, age, skin tone, and height) of subjects in Figure 17, from which we observe the following. First, airBP consistently achieves precise SBP and DBP estimations for various subjects of different gender, BMI, age, skin color, and height within MAE of 5 mmHg. Second, BMI bias has little impact on estimation accuracy, which meets the findings of existing research that the fat layer has little impact on the absorption rate of mmWave [64]. This shows that airBP can deliver robust BP monitoring performance for subjects with different levels of body fat. Third, estimation on female/older subjects is more accurate than on male/younger ones. We attribute this to the fact that thicker skin of males and more hydrated skin (which contains more water) of younger subjects [65] lead to more severe absorption of mmWave reflected by wrist arteries [66]. Fourth, it is interesting that estimation error increases for subjects with darker skin tone (i.e., Type V). Researchers have observed that dark skin tones absorb more reflection of blood pulsation for PPG sensors [67, 68], which makes the pulse sensing more challenging. We suspect that a similar phenomenon exists for mmWave.

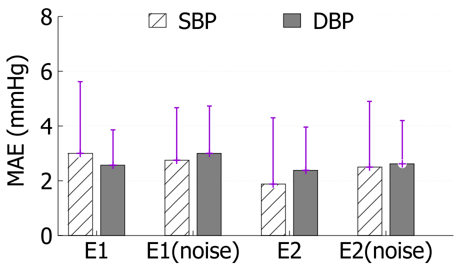

Fig. 18. Two unseen environments with clutter.

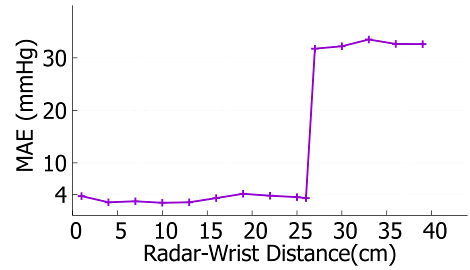

Fig. 19. SBP estimation error at various distances.

Impact of Environment. We further collect data from eight subjects under two unseen environments (E1: a dormitory, E2: a restaurant). In each environment, besides clean data collected without clutter around the device, we also collect data in noisy conditions, where the device is surrounded by clutter on a desk (i.e., computers, books, and cups). From the results in Figure 18, we observe that airBP performs robustly in each environment with MAE all within 4 mmHg, which indicates that airBP can focus on wrist artery motion and thus deliver *robust* BP estimation. It could reject the noise of surrounding clutter taking advantage of FMCW radar's nature [69] that could distinguish signals reflected from different distances.

Impact of Radar-Wrist Distance. To verify airBP's performance at different radar-wrist distances, we directly use the preceding model trained with the 12 cm dataset to estimate BP at other distances. Specifically, we separately change the distance from 1 cm to 39 cm by a step of 3 cm. At each distance, we record eight subjects' data and estimate BP using the preceding model (trained with the 12-cm dataset). From experimental results shown in Figure 19, airBP has an obvious effective distance range. To clarify it, we further collect extra data at distances of 25, 26, and 27 cm. From the preceding experiments, we observe the following. Our airBP remains effective at various distances within 26 cm, with MAE less than 4 mmHg. This indicates that airBP enables a larger effective range beyond the original data-collecting and training distance. Second, airBP totally fails when the radar-wrist distance exceeds 26 cm (27 cm, 30 cm, etc.). A further investigation finds that airBP loses the awareness of tiny arteries due to signal attenuation at such distances: MAE of pulse rate estimation increases from 2.30 bpm to more than 28 bpm. Although the effective range of 26 cm is enough for BP monitoring in working/sleeping scenarios (deployed on a desk or bed), we believe that the effective range could also be extended using mmWave sensors with a larger power. More details are discussed in Section 8. Third, it is interesting that airBP achieves the best performance at a middle distance (i.e., 4 to 16 cm), so we suggest implementing such a range as much as possible in practical usages.

Impact of Wrist Angle. Considering the user's daily wrist movements that may cause misalignment between it and the radar as shown in Figure 20(a), we evaluate the impact of wrist angle on airBP's performance. In particular, we examine airBP's performance from five subjects under different wrist angles (-60°, −30°, 0°, -30°, 60°) while keeping a radar-wrist distance of 12 cm. Leveraging the model trained with data collected under 0°, airBP's performance is illustrated in Figure 20(b). Our airBP achieves MAE of 4.60, 4.80, 3.42, 3.60, and 3.80 mmHg, respectively. It is interesting to observe that airBP performs well when the wrist angle increases from 0° to 60°, but it gets a little bit worse when the angle changes from 0° to −60°. We attribute this to the posture of the arm. When the forearm turns inward (surrounding the radar) as shown in the upper part of Figure 20(a), mmWave signals turn toward the wrist and the forearm, where the radial artery lies. When the forearm turns outward (shown in the bottom part of Figure 20(a)), more signals are reflected by the hand (where the artery buries inside the bone and muscles [70]) instead of the

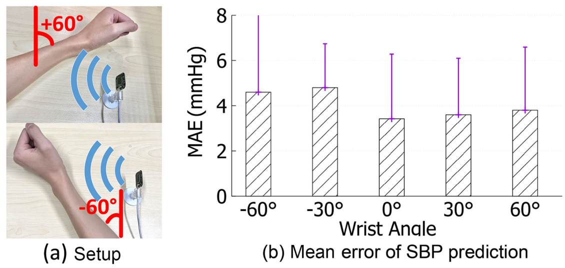

Fig. 20. Comparisons at different wrist angles.

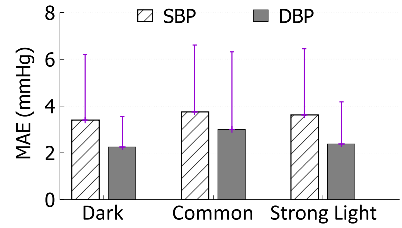

Fig. 21. Comparisons under different light conditions.

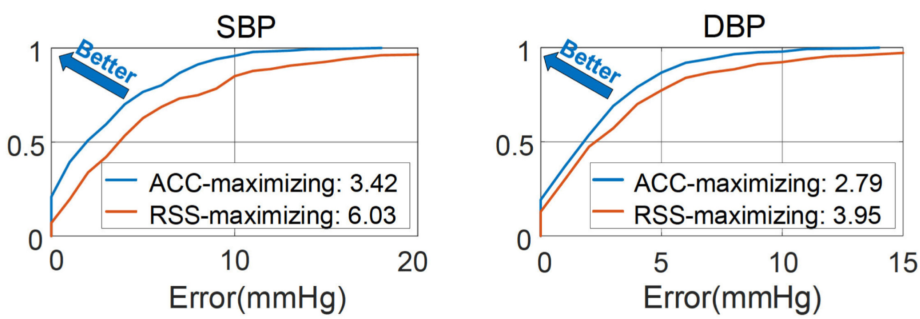

Fig. 22. Impact of different beam-forming methods on estimation of SBP (left) and DBP (right).

wrist. So in practical usages, we suggest using airBP with the forearm straight toward the radar or turning inward as shown in the upper part of Figure 20(a).

Impact of Ambient Light Conditions. As we expect, airBP is quite robust under different light conditions, as shown in Figure 21. It achieves MAE less than 4 mmHg for all three scenarios of dark, common light, and strong light. The result confirms airBP's advantages over optical sensor based methods, which suffer from the severe influence of ambient light conditions [21, 68].

7.3 Micro-Benchmark Experiments

Impact of ACC-Maximizing Beam-Forming. We further conduct experiments to compare ACC-maximizing with the traditional RSS-maximizing method [29], with other modules unchanged in airBP, and give the resulted MAE in Figure 22. We see that ACC-maximizing design reduces the MAE by 43.3% and 29.4% for SBP and DBP, respectively. As we discussed in Section 4, ACC-maximizing can help to locate the wrist artery pulse, and now we see that such capacity indeed leads to more accurate BP estimation.

Impact of the BP-Aware Pre-Training Method. We compare airBP's pre-training gains against two approaches: no-pre-training and vanilla pre-training using the original signal as a supervision label. From the results in Figure 23, we can observe the following. First, under no-pre-training, the MAE of estimated SBP and DBP increases to 2.63 and 3.07 times. In experiments, we find that without pre-training, the training loss still drops to a similar level, which indicates similar performance on the training set. However, the performance on the test set differs significantly (i.e., 8.98 mmHg vs. 3.42 mmHg on MAE), indicating that overfitting occurs without pre-training [71]. Meanwhile, an in-depth investigation is conducted to further explore the impact of BP-aware pre-training. We separately visualize the model's attention on the pulse wave samples of the test set with and without pre-training using the training set samples. Specifically, we upsample and normalize features

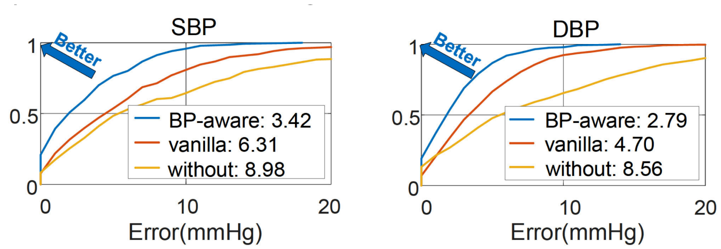

Fig. 23. Performance with different pre-training methods on estimation of SBP (left) and DBP (right).

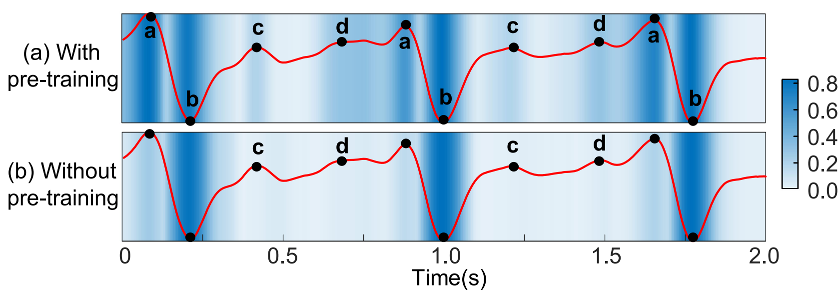

Fig. 24. Visualization example of the data-driven model's attention on the pulse wave (red line) in the test set with BP-aware pre-training using samples of the training set (top) and without pre-training (bottom). Darker color indicates stronger attention on the area. Without pre-training, the model loses the awareness of some important BP-related data points (i.e., points c and d).

extracted by the PreBlock in the encoder and present results in Figure 24. With BP-aware pre-training using the training set samples, our model learns to focus on the BP-related data points on the pulse wave of the test set—that is, samples around points *a*,*b*,*c*,*d* in Figure 24(a) have stronger attention. Without pre-training, the model fails to be aware of some important BP-related data points on the pulse wave—that is, the attention around points *c* and *d* in Figure 24(b) is not salient. Thus, it achieves worse BP estimation accuracy compared to BP-aware pre-training. Second, compared with the vanilla pre-training method, airBP reduces MAE of SBP and DBP by 45.8% and 40.6%, respectively. This verifies that airBP's pre-training can effectively handle the pulse sparsity and successfully extract BP features.

Impact of the Convolutional Transformer. In Figure 25, we compare airBP's convolutional transformer design against BLSTM [47] and the traditional transformer [50]. Training with the same schedule including BP-aware pre-training, we can observe the following. First, BLSTM [47] achieves the worst performance with MAE of 8.26 mmHg and 5.68 mmHg for SBP and DBP estimation, respectively, which is more than two times that of ours. BLSTM fails to capture rather sparse BP-related features from long-term time-series data. Second, leveraging the global attention mechanism, the transformer achieves better performance compared with BLSTM, resulting in a reduction of 40% and 35% MAE for SBP and DBP estimation. However, airBP achieves the best accuracy with MAE of 3.42 mmHg and 2.79 mmHg for SBP and DBP, respectively, which further reduces MAE by 30% and 25% over the transformer. The results confirm the advantages of airBP's elaborate convolutional transformer, which better captures the periodically occurred BP-related features leveraging the local contextual attention.

Impact of the Custom-Designed Feature Encoder. We also compare the design choice of the encoder with the replacement of two state-of-the-art deep learning models: DNN [54] and

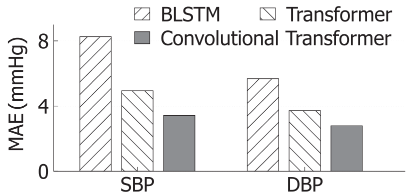

Fig. 25. Impact of transformer design.

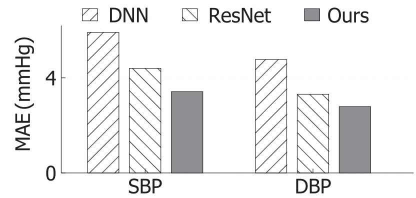

Fig. 26. Impact of encoder choice.

34-layer ResNet [72]. As shown in Figure 26, the MAE of their SBP estimation is 5.91 and 4.40 mmHg, and that of DBP is 4.77 and 3.31 mmHg, respectively. We can observe the following. First, compared to DNN, ResNet achieves lower errors due to its deep-layer design, which indicates that a complex deep learning model is more suitable for long-term arterial pulses in BP estimation. Second, compared to ResNet, airBP achieves a remarkable reduction by more than 22.3% and 15.7% on MAE for SBP and DBP, respectively. This confirms the effectiveness of applying the attention mechanism to focus on the most informative BP features among long-series data.

8 DISCUSSION AND FUTURE WORK

Extending BP Monitoring Range. Our airBP's performance drops when the radar-wrist distance exceeds the range of 26 cm. The main reason for the performance drop at a longer distance is the severe signal attenuation that makes it harder to capture the pulse wave. In this case, the ACC-maximizing beam-forming also fails to focus on arterial activity. In the future, we envision room-scale BP monitoring, which requires a monitoring range of up to a few meters. We plan to try this by employing an mmWave radar with larger transmitting power, such as TI AWR2243 [73], to improve the signal-to-noise ratio at longer distances.

BP Monitoring for Mobile Users. It would be more useful if we could monitor BP while users are performing daily activities, such as reading or watching TV, without requiring them to put an arm on a specific device and remain still. Toward this goal, we need to combine user behavior sensing with airBP, which can identify the moving status of the user, and eliminate the RSS change caused by other body part motions. Moreover, we also need an agile target tracking technique to keep focusing on the wrist. We leave these for future exploration.

BP Monitoring with Arterial Pulse Wave Recorded for a Shorter Time. Currently, airBP takes the arterial pulse wave recorded for 35 seconds (usually maintains more than 35 pulse periods) as input to perform accurate BP estimation. In the future, we plan to move forward achieving BP monitoring with the arterial pulse wave recorded for a shorter time, such as 3 seconds. For practical usage, the requirements of a shorter pulse waveform will not only increase the speed of BP measurement but also enable more user freedom. However, in experiments taking data of 3 seconds as input, we find that the MAE of SBP estimation increases to 7.67 mmHg, which is more than two times that (3.42 mmHg) using 35 seconds of data. In future work, we will attempt to achieve a better balance between accuracy and requirements of pulse wavelength.

9 RELATED WORK

In this section, we introduce the research most related to our work, including existing BP monitoring methods, mmWave sensing, and sparse feature extraction via data-driven methods.

9.1 Existing BP Monitoring Methods

BP monitoring methods can be classified into three categories according to the portability of devices: cuff-based methods, wearable device based methods, and contactless methods.

Cuff-Based Methods. Traditionally, an inflatable cuff and a stethoscope are used to capture the pulse sound of the brachial artery while successively compressing one's upper arm by inflating and deflating, which is usually conducted by professional medical staff. When the sound is observed, BP values are obtained according to the corresponding cuff pressure [14]. For non-clinic BP measurement, digital BP devices are suggested for daily usage, which applies oscillometric methods to obtain BP [74]. Although providing reliable results, they are bulky with poor portability, which hinders comfortable and convenient monitoring.

Wearable-Based Methods. With the development of PPG sensors [75], much research effort is devoted to cuffless BP monitoring with PPG in wearable devices for portable and convenient BP monitoring. In particular, Seismo [19] and SeimoWatch [20] calculate the PTT from the heart to the wrist/finger using a PPG/camera and embedded accelerometer, respectively. They require *user effort* to press the device on the chest and simultaneously press his/her's figure, so as to separately obtain the time of aortic valve opening and pulse arrival. Naptics [76] and Glabella [77] attach multiple PPGs in tight shorts/glasses to calculate PTT between different body points. However, they are only applicable in limited scenarios—that is, glasses and tight shorts are not suitable while sleeping and working, respectively. Recently, eBP [16] has further incorporated in-ear PPG and a digital pump that periodically compresses blood vessels in the ear. LiveOne [17] proposes wrist-worn devices with pressure sensors that closely compress the skin to measure BP. Yet, the unpleasant and even injurious [18] compression process concerns users who require frequent BP monitoring. The most recent work, named *Crisp-BP* [21], enables BP measurement with PPG embedded in wrist-worn devices, based on the RWTT principle. Although providing continuous BP monitoring, the strict requirements of sensor-skin contact pressure bring an unpleasant user experience of carefully tuning—that is, either higher or lower pressure leads to poor accuracy of BP estimation. In contrast, the fundamental difference of airBP lies in that it is a contactless solution that enables BP monitoring without wearing any device.

Contactless Methods. Some most recent work explores the feasibility of contactless BP monitoring [25, 78–81]. Although promising, these works are limited to a feasibility study and are far from a working system. Generally, they could be classified into two categories based on sensing targets. The first is the chest. Some researchers [78–80] employ mmWave radar to measure the chest movement associated with heart activity and further predict BP based on the extracted features (e.g., the time interval between two heart sounds). They require subjects to hold their breath for 20 seconds to eliminate the impact of respiration and reported performance on *a limited number of subjects* (i.e., 6∽8 subjects). The second is the wrist. Compared to the chest, wrist artery sensing is less affected by the noise of respiration and thus is more promising as the sensing region for BP monitoring. However, the existing wrist-sensing methods are still far from practical due to the unsolved challenge in sensing tiny wrist arteries. Recent research [81] employs a 24-GHz mmWave radar deployed 4 cm away from the wrist to obtain the RWTT (named *R-PTT* in that work) maintained in the wrist pulse waveform and further validates its correlation on a single person with only eight measurements. Another recent work [25] develops a wearable BP monitor using a 60-GHz mmWave radar to sense the radial artery at 3 mm above and maps the pulse waveform into BP units. Among the collected 20,213 samples, only 1,096 samples (5% of the dataset) with a high signal-to-noise ratio could be employed to estimate BP. Meanwhile, extra calibration using a cuff-based BP monitor is required for each subject, which hinders its practical application. Different from these existing methods, we not only propose a fundamental contactless wrist artery sensing model based on the Lambert-Beer law but also design an ACC-maximizing beam-forming method and a custom-designed deep learning pipeline to enable robust and accurate BP estimation for multiple subjects without extra calibration. The designs were filed as two patents in early 2022 [82, 83].

9.2 mmWave Sensing

Due to the short wavelength, mmWave enables high-resolution and fine-grained motion caption. Hence, it has a wide range of applications, such as machine vibration measurement [4], human activity recognition [6, 7], and vital sign monitoring [8–10, 84]. Some most recent work attempts to employ mmWave to capture some more fine-grained vital signs (i.e., SCG and ECG) [3, 5, 11]. These methods utilize phase variation to capture cardiac activity from the human chest, which has a larger reflection area and distention amplitude than the wrist. However, airBP explores the feasibility of a more challenging task: estimating BP using more fine-grained cardiac activity from more tiny wrist arteries.

9.3 Sparse Feature Extraction Via Data-Driven Methods

Recently, the deep learning model has enabled automatic feature extraction from noisy data, such as image [85, 86] and speech [87, 88]. Different from content-rich media like the image or speech, time series of radio frequency signals contain rather sparse features [3, 5, 11, 47]. To capture the long-term dependencies in time-series data, BLSTM [21, 47] is widely employed, but it fails to capture long dependencies [51]. Hence, researchers employ more complex deep learning models [50, 54, 72], among which the transformer [50] shows its advantages in capturing recurring patterns in long series, using the global attention mechanism. In addition, the convolutional transformer [89] further enhances the context awareness. Inspired by this progress, airBP customizes a multi-attention neural network model combined with the BP-aware pre-training so as to extract sparse BP-related features from noisy mmWave RSS variation.

10 CONCLUSION

In this article, we designed and built a new BP monitoring system called *airBP*, which enables precise and robust BP estimation leveraging 60-GHz mmWave wireless sensing. To realize airBP, we proposed ACC-maximizing beam-forming to focus on the tiny motion of wrist arteries, and customized/designed a multi-attention deep learning architecture to extract effective BP features from sparse and noisy mmWave signals. Evaluation on 41 subjects showed that airBP's BP estimation meets the FDA's AAMI standard. In other words, airBP pushes the mmWave sensing ability to the minute wrist artery motion and validates BP monitoring in a contactless way with extensive experiments.

APPENDIX

A BP ESTIMATION FROM THE ARTERIAL PULSE VIA RWTT

Medical research shows that the pulse wave in the wrist artery contains BP-related features (i.e. RWTT [41, 90, 91]), which is illustrated in Figure 27. In particular, as shown in Figure 27(a), when pulse waves (black arrow) travel to bifurcations of the arterial tree, part of them will be reflected (yellow arrow). Thus, at the same sensing point (the blue arrow in Figure 27(a)), we can sense twice pulsation during one pulse period, which is illustrated by the black and gray curve in Figure 27(b). The interval between their arrival time (i.e., t1 and t3 in Figure 27(b)), named *RWTT*, is inversely correlated with the travel speed of blood flow [92]. Since higher speed indicates higher BP [93], one can estimate BP from RWTT, denoted by T_{RWTT}, using

$$BP = \frac{\alpha}{T_{RWTT}} + \mu, \tag{20}$$

where α and μ are user-specific parameters [94].

To examine whether we could precisely extract RWTT from a pulse wave generated by mmWave and use Equation (20) to directly compute BP, we collect 43 samples from one subject and record the

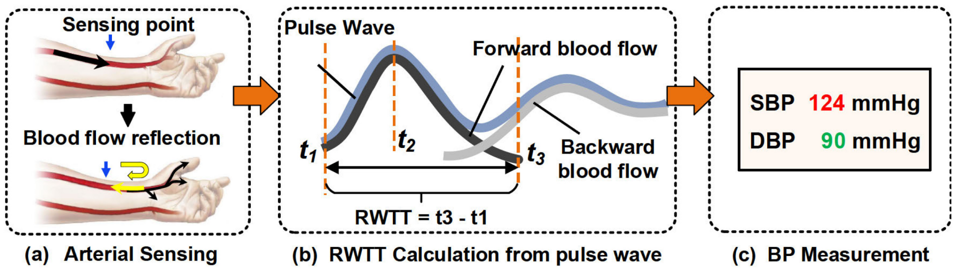

Fig. 27. Overview of BP measurement from the arterial pulse based on RWTT.

ground-truth BP using a digital BP device. Results are that the obtained $\frac{1}{T_{RWTT}}$ is of low correlation with BP ground truth, which is only 0.3092. In contrast, the correlation is more than 0.82 when using contact or intrusive sensors [21]. This verifies the challenges to precisely capture the fine-grained details of pulse waves using mmWave from a certain distance. In addition, this motivates our new designs in Section 5 to handle the ambiguous nature of pulse waveforms generated by contactless sensing.

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